Midterm 2 Prep

CPSC 330 Tutorial 7

Midterm Format

- Some multiple choice/short answer, some workspace questions (in separate links)
- 50 minutes!!
  - Open workspaces first!!!
- Save OFTEN
  - How to save workspace? Demo!
- Up to and including recommendation systems, cumulative

How to Study?

- Try PL at some point
- Study iClicker (short-answer)
- Study common coding things you see a lot (workspace)
- Unrealistic to remember every bit of syntax - get comfortable navigating course notes and documentation
    - Remembering which page in the course notes different topics are on really helps!
- Look at workspace details before exam (see Ed)

Kahoot time :)

Ensembles

Which statement about the source(s) of randomness in random forests is correct?

A) Each tree is built using the same random subset of features
B) Each tree is built using a random sample drawn with replacement from the training set
C)For numerical features, the threshold for each node is picked at random
D) All of the above

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Feature importances

You are training a random forest classifier and want to know which feature pushes its predictions furthest in the positive direction and which feature pushes its predictions furthest in the negative direction. You should:

A) Look at the magnitude and sign of its coefficients
B) Use sklearn's feature_importances_
C) Use sklearn's permutation_importances
D) Create a SHAP summary plot

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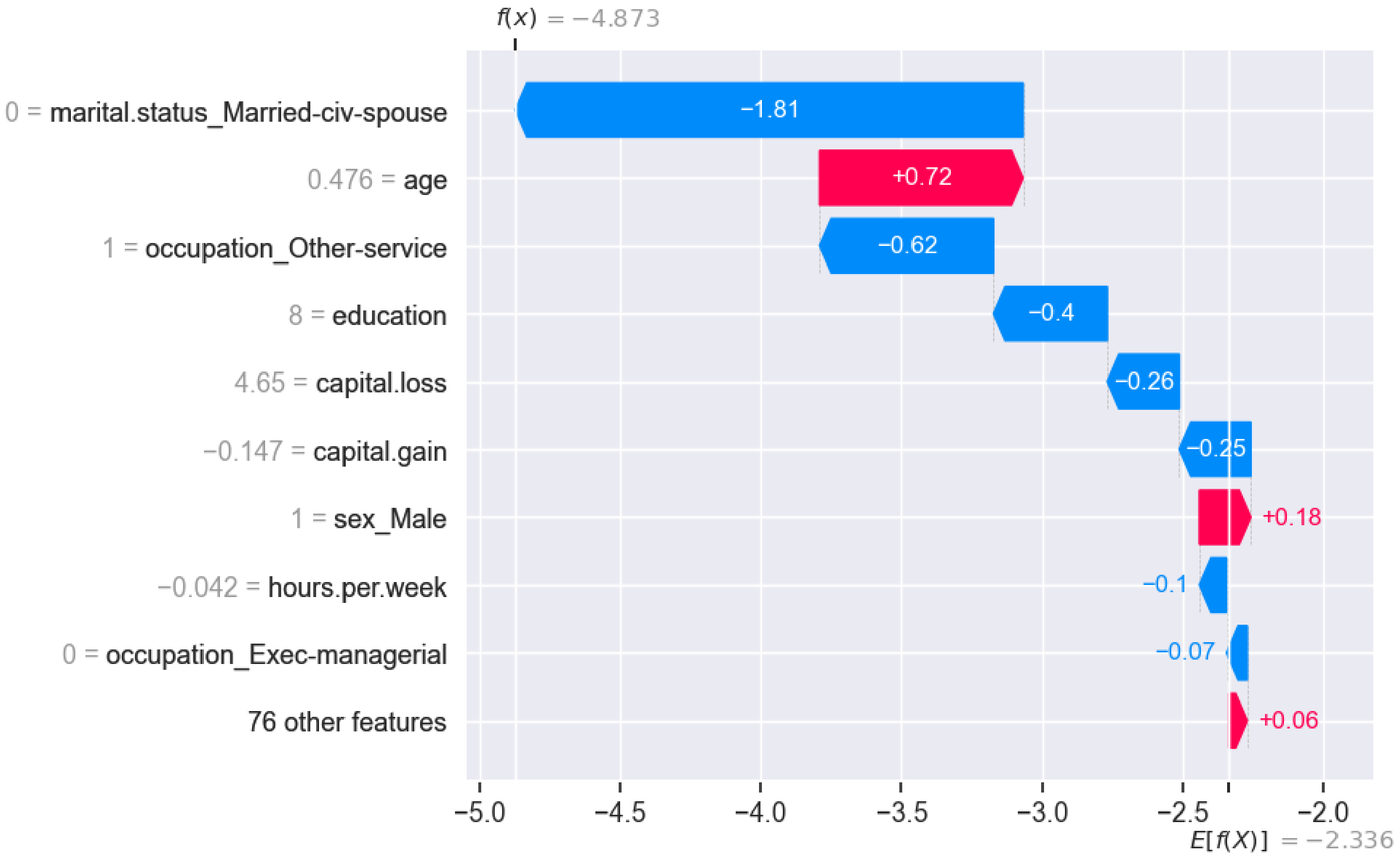
SHAP

X = this person not being married is pushing their prediction towards the negatives

Y = this person's age is below the average for the dataset

Which of the below is true? Assume StandardScaler is used.

A) X and Y
B) !X and Y
C) X and !Y
D) !X and !Y



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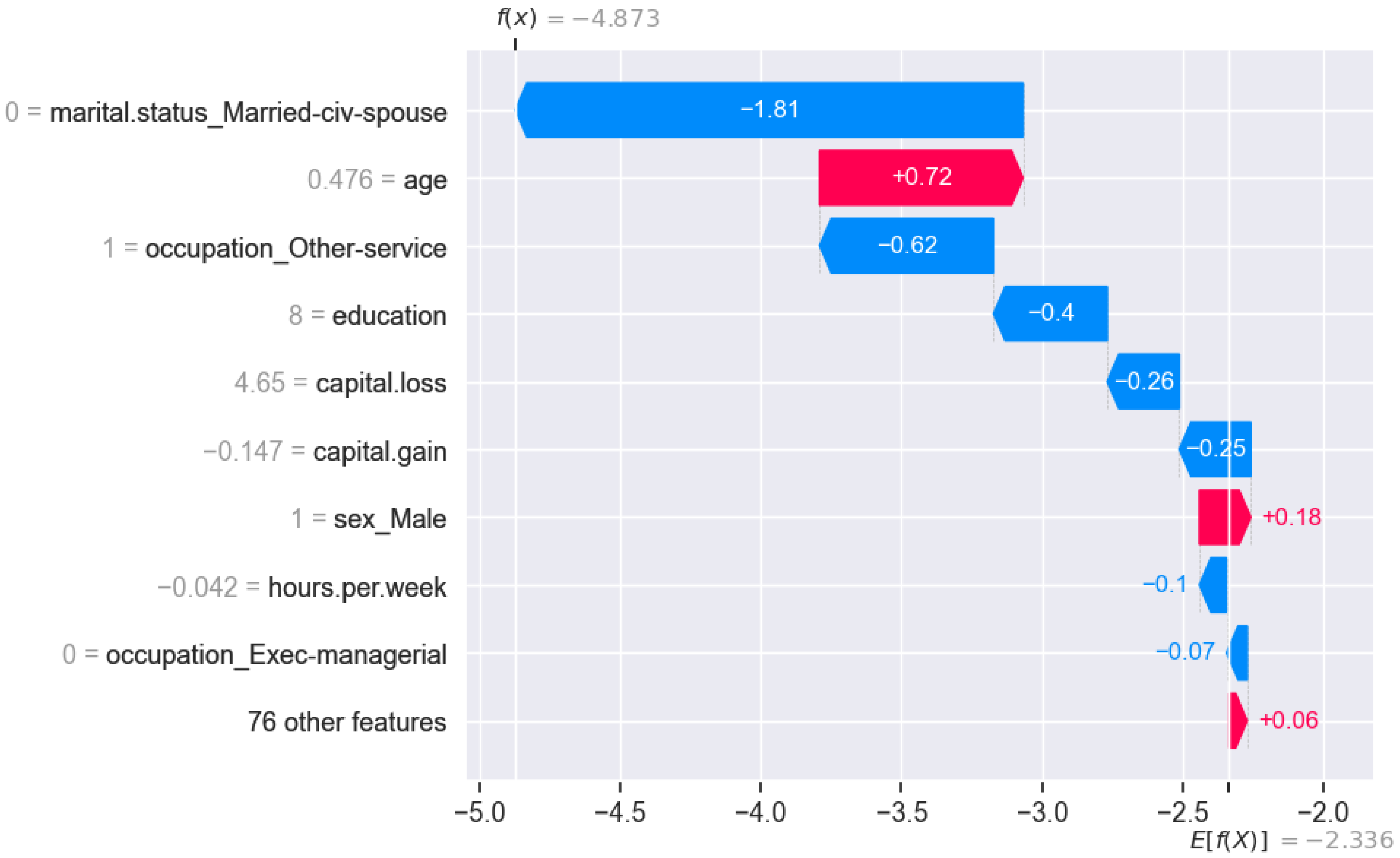

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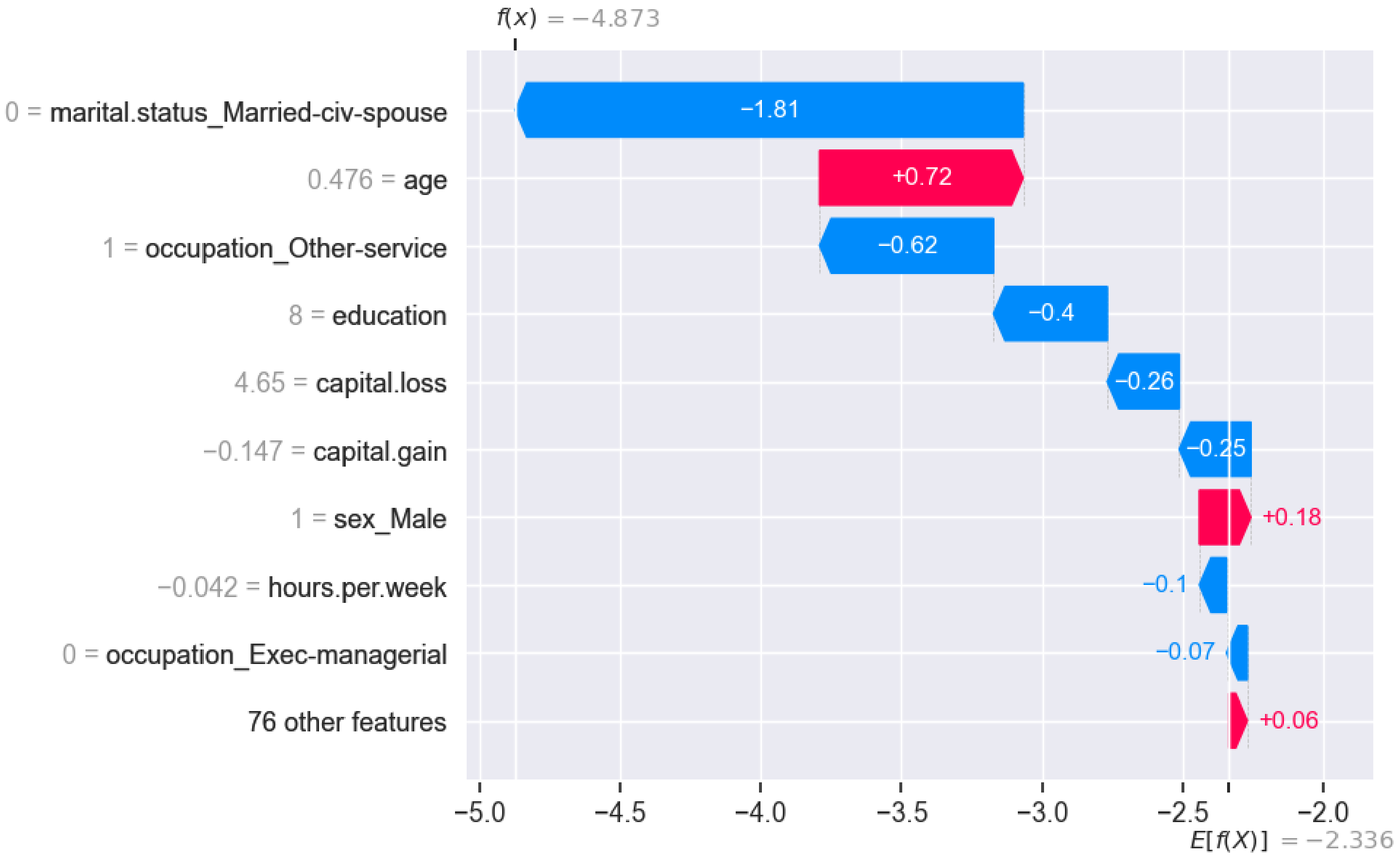
Does this dataset have more samples from the positive or negative class in it?

A) Positive
B) Negative
C) Can't tell from the SHAP plot



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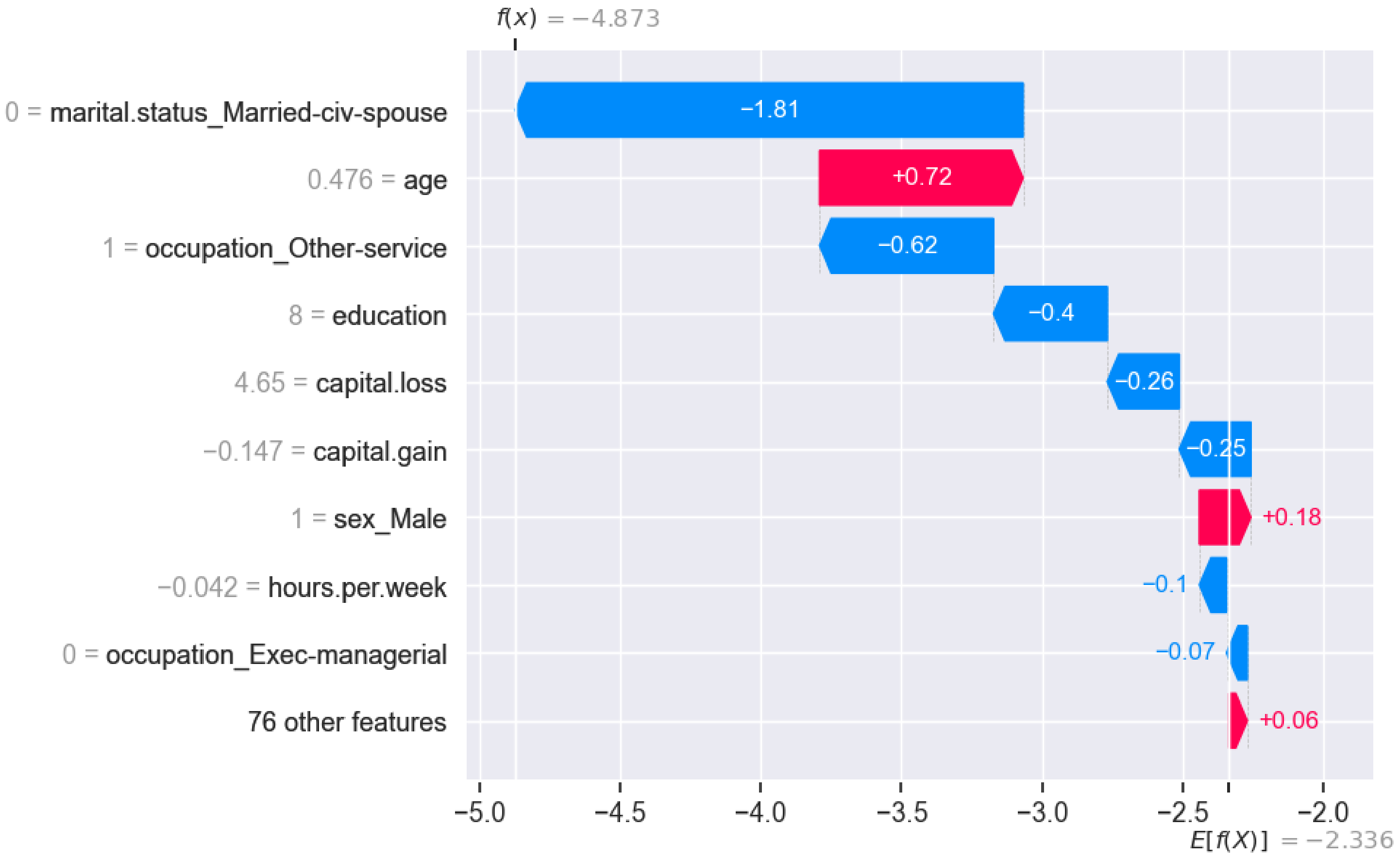

Features

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A) Performing feature selection using kNN
B) Performing feature selection using logistic regression
C) Reducing k
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K-means

Which statement about K-means is true?

A) It is useful to use both the Elbow method and the Silhouette method to pick K
B) K-means always converges to the same clusters
C) K-means is guaranteed to find the optimal solution
D) K-means does not always assign a cluster to each point

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DBSCAN

Which statement about DBSCAN is false?

A) We do not have to specify the number of clusters
B) DBSCAN can assign a cluster to new data points
C) DBSCAN can capture complex cluster shapes
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Clustering

Given the same data, K-means and DBSCAN are most likely to produce similar results if:

A) K is very high, epsilon is very high, min_samples is very high
B) K is very low, epsilon is very low, min_samples is very high
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Recommender systems

How does content-based filtering predict a user's ratings for a given movie?

A) It creates a regression model for the movie that predicts a rating based on the user's features
B) It averages all ratings for that movie
C) It averages the ratings that similar users gave to that movie
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Recommender systems

A video platform notices that users who watch documentaries about climate science are repeatedly recommended very similar documentaries.

Users rarely receive recommendations for related but different topics such as environmental economics or sustainability policy.

Which change would most likely increase recommendation diversity?

A) Increasing the weight of item metadata in the content-based model
B) Switching to collaborative filtering or a hybrid approach
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Good Luck on MT2! :)